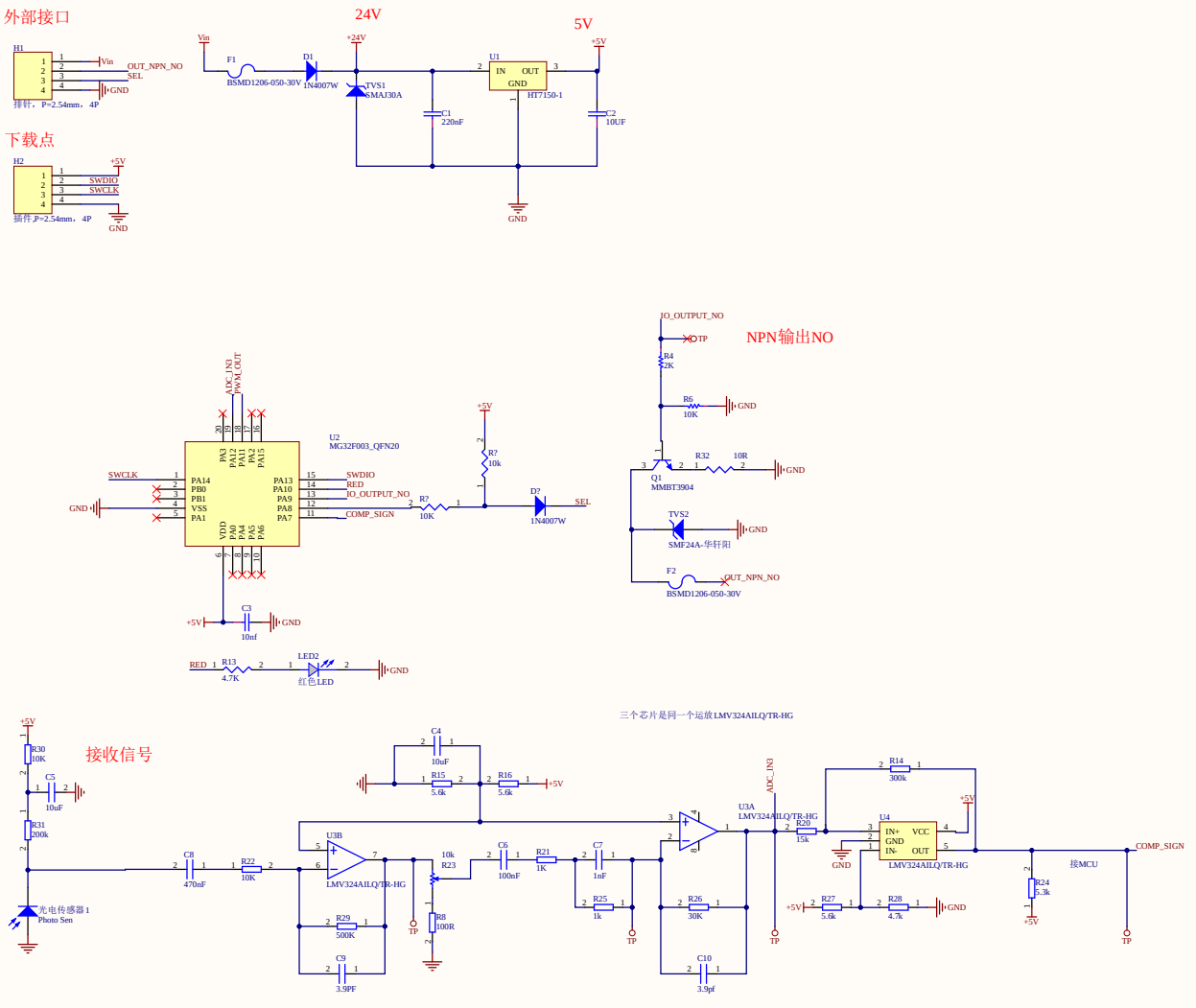

外部接口
H1
Vin
OUT_NPN_NO
SEL
GND
排针，P=2.54mm，4P
下载点
H2
+5V
SWDIO
SWCLK
GND
插件,P=2.54mm，4P
24V
5V
Vin
F1
BSMD1206-050-30V
D1
1N4007W
+24V
TVS1
SMAJ30A
C1
220nF
U1
IN
OUT
GND
HT7150-1
+5V
C2
10UF
GND
ADC_IN3
PWM_OUT
U2
MG32F003_QFN20
SWCLK
GND
SWDIO
RED
IO_OUTPUT_NO
COMP_SIGN
R7
10k
D2
1N4007W
SEL
C3
10nf
RED
R13
4.7K
LED2
红色LED
IO_OUTPUT_NO
TP
NPN输出NO
R4
22k
R6
10K
R32
10R
Q1
MMBT3904
TVS2
SMF24A-华轩阳
F2
BSMD1206-050-30V
OUT_NPN_NO
三个芯片是同一个运放LMV324AILQ/TR-HG
接收信号
R30
10K
C5
10uF
R31
200k
光电传感器1
Photo Sen
C8
470nF
R22
10K
U3B
LMV324AILQ/TR-HG
C4
10uF
R15
5.6k
R16
5.6k
10k
R23
R8
100R
C6
100nF
R21
1K
C7
1nF
R25
1k
R29
500K
C9
3.9PF
U3A
LMV324AILQ/TR-HG
R20
15k
R26
30K
C10
3.9pf
U4
IN+
GND
IN-
VCC
OUT
LMV324AILQ/TR-HG
R14
300k
R27
5.6k
R28
4.7k
R24
5.3k
接MCU
COMP_SIGN

Comment	Description	Designator	Footprint	LibRef	Quantity